Student Lifestyle & Academic Performance

Analysis Dashboard

Exploring the relationship between student lifestyle habits and academic performance across 2,000 students.

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Project at a Glance

PROJECT OBJECTIVE

Investigate which student lifestyle habits most strongly influence academic performance (GPA), and by how much — using Excel, Power Query, and interactive dashboard design.

DATASET SCOPE

- 2,000 synthetic university students
- 8 original columns · 0 missing values
- 5 lifestyle activity columns · GPA · Stress Level
- 4 engineered columns via Power Query

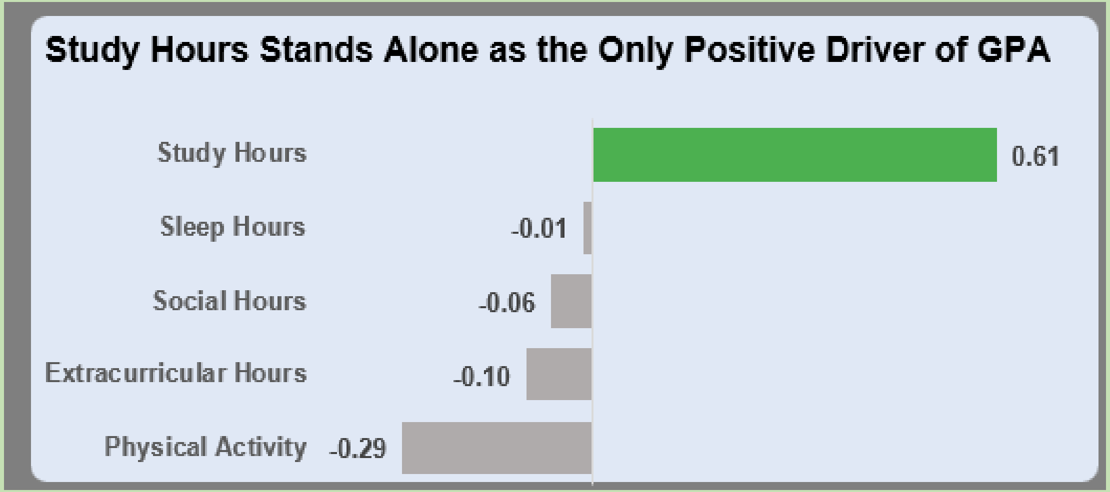
Full analysis and working file available at: [View Full Work Here]

Dashboard Overview

Fully interactive · Three slicers · Five KPI cards · Eight chart panels

Interactive Dashboard · Excel 2019 · Power Query · Slicers

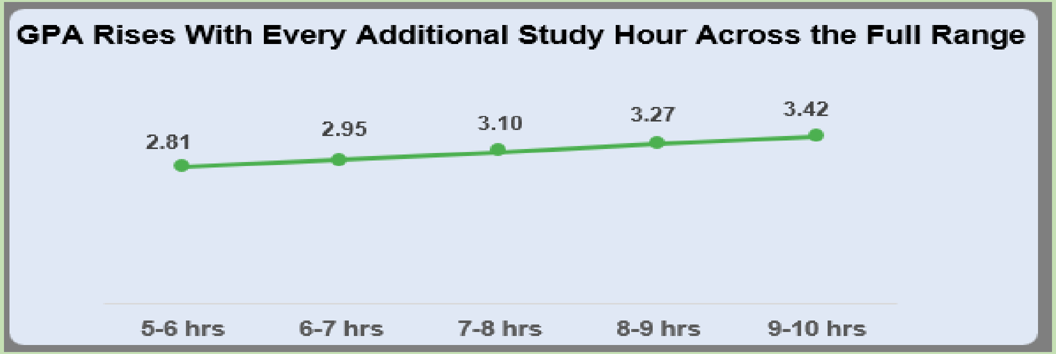
GPA distribution across activities



KEY FINDINGS

- Study Hours is the ONLY activity with a positive GPA impact (+0.61 pts).
- Every other activity — Physical Activity, Social, Extracurricular, Sleep — shows a neutral or negative net effect.
- Academic performance is not driven by balance or variety of activities.
- It is driven by deliberate concentration of time on study.

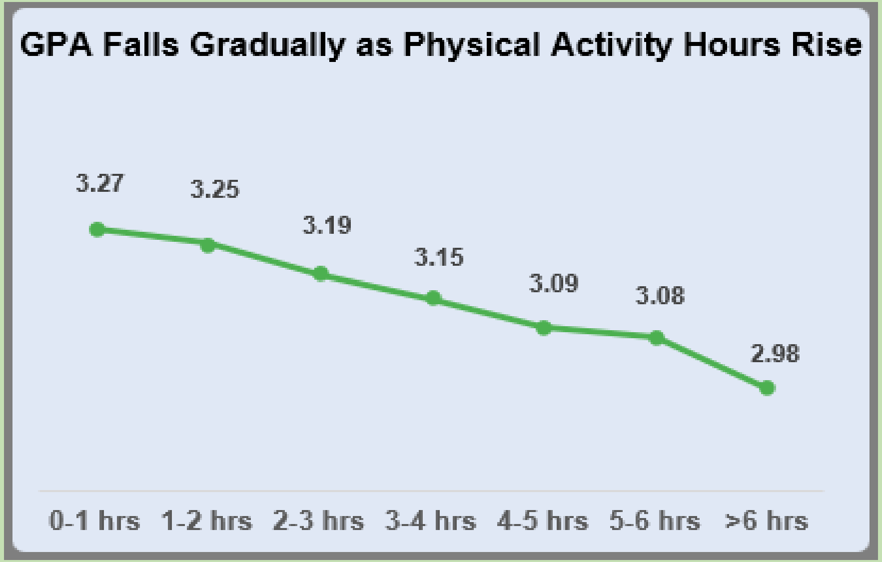
GPA Vs Study Hour



KEY FINDINGS

- GPA rises with every additional study hour — no exceptions across the 5–10 hr range.
- Average GPA climbs from 2.81 (5–6 hrs) to 3.42 (9–10 hours).
- The relationship is linear and consistent — no plateau, no threshold, no diminishing returns within the observed range.

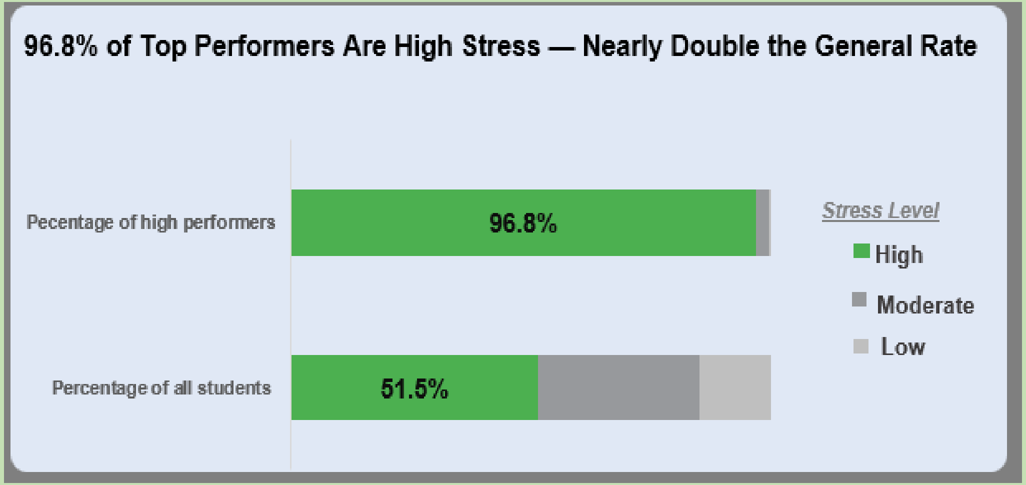
Physical Activity & GPA



KEY FINDINGS

- GPA falls gradually as physical activity hours increase — from 3.27 (0–1 hr) to 2.98 (>6 hrs).
- The decline is slow and progressive, not sudden.
- Physical activity is not academically harmful. It competes with finite time. Every hour exercising is an hour not studying — and study hours drive GPA.

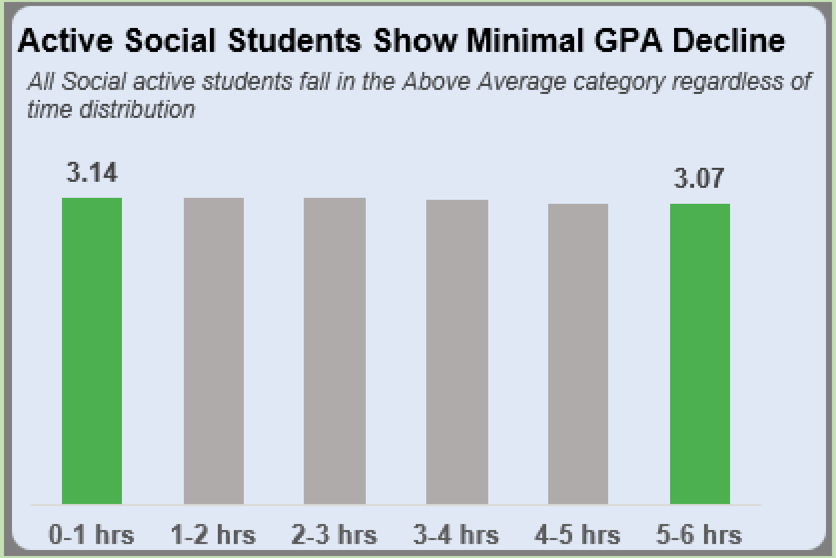
Stress Profile of High Performers



KEY FINDINGS

- 96.8% of High Performers (GPA ≥ 3.5) experience High Stress.
- Across all students, only 51.5% are High Stress — nearly half the rate.
- High stress is a feature of top academic performance in this dataset. Academic excellence appears to come with a consistent pressure cost.

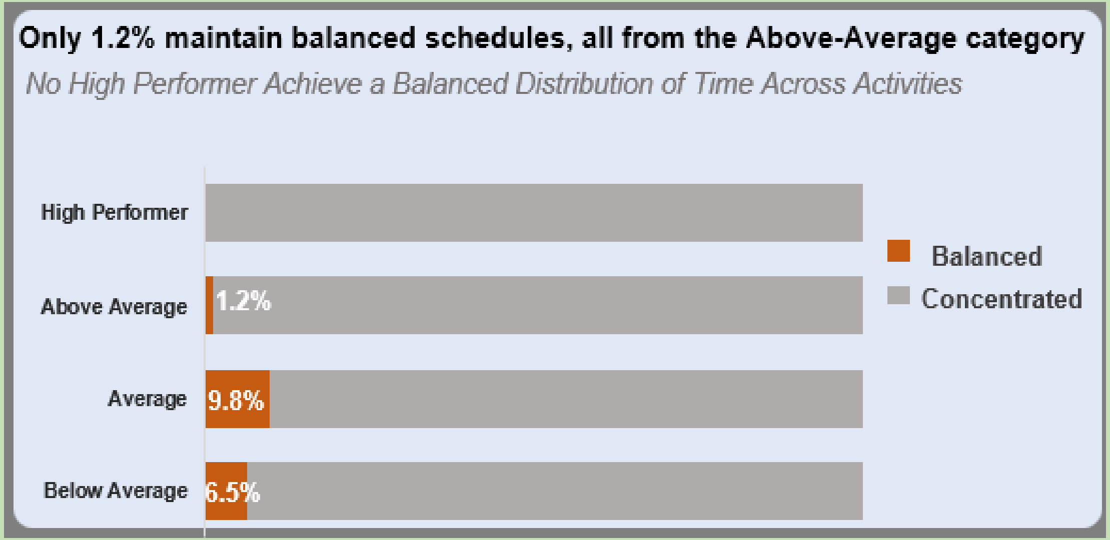
Social Activity & GPA



KEY FINDINGS

- Socially active students remain Above Average in GPA regardless of social hours.
- Total GPA decline across the full social hours range is just 0.07 points.
- The cultural assumption that socializing damages grades is not supported by this data. Students can maintain strong performance while remaining socially active.

Does the Balanced Student Exist?



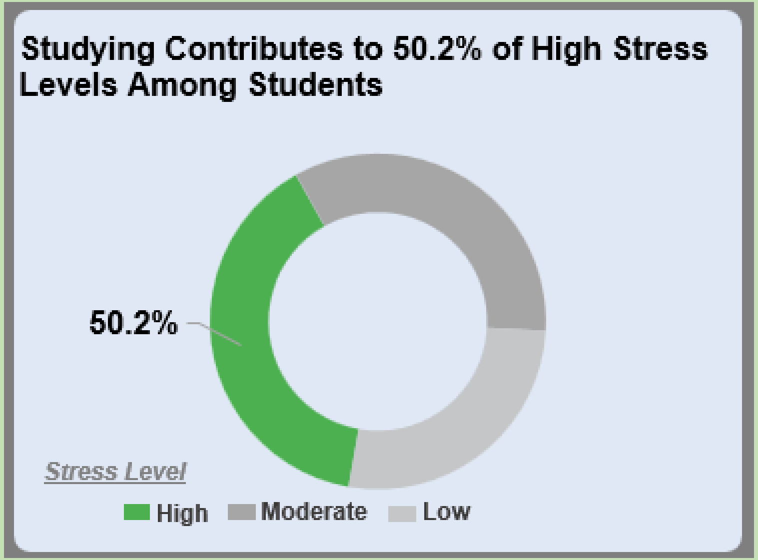
METHODOLOGY

- Balance defined as Activity StdDev ≤ 1.5 across Study, Social, Physical Activity & Extracurricular hours per student.

KEY FINDINGS

- **0% of High Performers achieve a genuinely balanced lifestyle.**
- **Only 1.2% of Above Average students meet the balance criteria.**
- **Below Average students show the highest balance rate (6.5%).**
- **True lifestyle balance and peak academic performance are mutually exclusive in this dataset. Top performance requires deliberate concentration, not even distribution.**

What Drives Student Stress?



KEY FINDINGS

- High Stress students dedicate 50.2% of productive hours to study vs 34.6% for Low Stress.
- High Stress student study ~3 more hours/day than Low Stress peers.
- Stress is driven by study concentration, not by juggling many activities. The allocation of time — not the total — is what separates stress levels.

Analytical Skills Demonstrated

Techniques and reasoning applied throughout this project

Data Transformation

Power Query: loading, type-setting, multi-step transformation pipeline

Chart Selection

Diverging bar, line, 100% stacked bar, clustered bar — matched to analytical question

Feature Engineering

5 meaningful columns derived from raw data with documented rationale

Dashboard Design

Single-screen interactive dashboard with 3 slicers, 5 KPI cards, static + dynamic charts

Statistical Reasoning

Standard deviation applied as a behavioural profiling metric; threshold validated against raw rows

Critical Validation

Independently identified misleading axis scaling, flawed threshold, and chart type mismatch

VBA Automation

Macro recorded to automate filter clearing and improve dashboard usability

Tools Used

Technical stack powering this end-to-end analysis

Microsoft Excel

Power Query

VBA Macro

View the full project on GitHub: https://github.com/Princess-Okere/Student-Lifestyle-and-Academic-Performance-Analysis

Includes: Working Excel file | Full analysis | Interactive dashboard | README documentation

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